

Exploratory Data Analysis

Auto-MPG dataset



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ANALYSIS GOALS

- *How does car origin relate to car characteristics?*
- *How does car origin influence characteristics evolution over time?*

AUTO-MPG DATASET

	mpg	cylinders	displacement	horsepower	weight	acceleration	model_year	origin	car_name
0	18.0	8	307.0	130	3504	12.0	70	1	chevrolet chevelle malibu
1	15.0	8	350.0	165	3693	11.5	70	1	buick skylark 320
2	18.0	8	318.0	150	3436	11.0	70	1	plymouth satellite
3	16.0	8	304.0	150	3433	12.0	70	1	amc rebel sst
4	17.0	8	302.0	140	3449	10.5	70	1	ford torino

Shape: 398 rows and 9 columns.

Variables

- **Categorical:** origin, cylinders, and car_name;
- **Continuous:** mpg, horsepower, weight, displacement, acceleration, and model_year.

DATA CLEANING

Original dataframe

- Null values: 0 in all columns
- NaN: 0 in all columns

#	Column	Dtype
0	mpg	float64
1	cylinders	int64
2	displacement	float64
3	horsepower	object
4	weight	int64
5	acceleration	float64
6	model_year	int64
7	origin	int64
8	car_name	object
dtypes:	float64(3), int64(4), object(2)	

Conversion of *horsepower* to numeric

Conversion of *horsepower* to numeric

- Null values: 0 in all columns
- NaN: 6 in *horsepower* and 0 in the remaining columns

#	Column	Dtype
0	mpg	float64
1	cylinders	int64
2	displacement	float64
3	horsepower	float64
4	weight	int64
5	acceleration	float64
6	model_year	int64
7	origin	int64
8	car_name	object
dtypes:	float64(4), int64(4), object(2)	

NAN VALUES

	cylinders	displacement	horsepower	model_year	car_name
32	4	98.0	NaN	71	ford pinto
126	6	200.0	NaN	74	ford maverick
330	4	85.0	NaN	80	renault lecar deluxe
336	4	140.0	NaN	80	ford mustang cobra
354	4	100.0	NaN	81	renault 18i
374	4	151.0	NaN	82	amc concord dl

NAN VALUES

	cylinders	displacement	horsepower	model_year	car_name
32	4	98.0	NaN	71	ford pinto

	cylinders	displacement	horsepower	model_year	car_name
32	4	98.0	NaN	71	ford pinto
61	4	122.0	86.0	72	ford pinto runabout
80	4	122.0	86.0	72	ford pinto (sw)
112	4	122.0	85.0	73	ford pinto
130	4	122.0	80.0	74	ford pinto
168	4	140.0	83.0	75	ford pinto
174	6	171.0	97.0	75	ford pinto
206	4	140.0	72.0	76	ford pinto

NaN replaced with average

NAN VALUES

	cylinders	displacement	horsepower	model_year	car_name
126	6	200.0	NaN	74	ford maverick

	cylinders	displacement	horsepower	model_year	car_name
17	6	200.0	85.0	70	ford maverick
100	6	250.0	88.0	73	ford maverick
126	6	200.0	NaN	74	ford maverick
155	6	250.0	72.0	75	ford maverick
193	6	200.0	81.0	76	ford maverick

NaN replaced with average

NAN VALUES

	cylinders	displacement	horsepower	model_year	car_name
330	4	85.0	NaN	80	renault lecar deluxe
354	4	100.0	NaN	81	renault 18i

	cylinders	displacement	horsepower	model_year	car_name
79	4	96.0	69.0	72	renault 12 (sw)
186	4	101.0	83.0	76	renault 12tl
218	4	79.0	58.0	77	renault 5 gtl
330	4	85.0	NaN	80	renault lecar deluxe
354	4	100.0	NaN	81	renault 18i

NaN values dropped

NAN VALUES

	cylinders	displacement	horsepower	model_year	car_name
336	4	140.0	NaN	80	ford mustang cobra

	cylinders	displacement	horsepower	model_year	car_name
48	6	250.0	88.0	71	ford mustang
166	8	302.0	129.0	75	ford mustang ii
236	4	140.0	89.0	77	ford mustang ii 2+2
336	4	140.0	NaN	80	ford mustang cobra
393	4	140.0	86.0	82	ford mustang gl

NaN replaced with average

NAN VALUES

	cylinders	displacement	horsepower	model_year	car_name
374	4	151.0	NaN	82	amc concord dl

	cylinders	displacement	horsepower	model_year	car_name
257	6	232.0	90.0	78	amc concord
261	6	258.0	120.0	78	amc concord d/l
283	6	232.0	90.0	79	amc concord dl 6
315	4	151.0	90.0	80	amc concord
374	4	151.0	NaN	82	amc concord dl

NaN replaced with base car model horsepower value

OUTLIERS

- Shape after dealing with NaN: 396 rows and 9 columns
- Outliers count per column Tukey's method (IQR)
 - mpg – 1 outlier
 - horsepower – 11 outliers → greater impact on average → drop
 - acceleration – 9 outliers
- After dropping horsepower outliers
 - mpg – 1 outlier
 - horsepower – 11 outliers
 - acceleration – 6 outliers

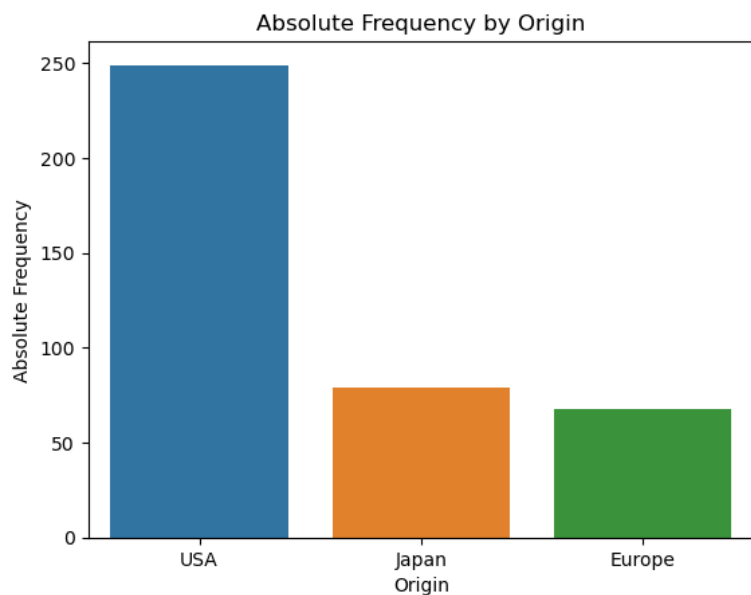


Dataframe after addressing
NaN values kept

EDA UNIVARIATE - CATEGORICAL

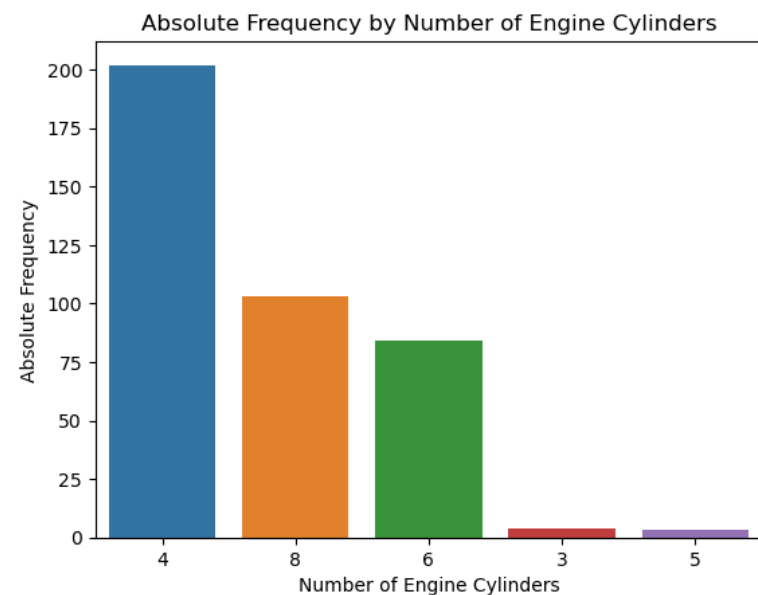
Origin

Origin	Absolute Frequency	Relative Frequency
USA	249	0.63
Japan	79	0.20
Europe	68	0.17



Cylinders

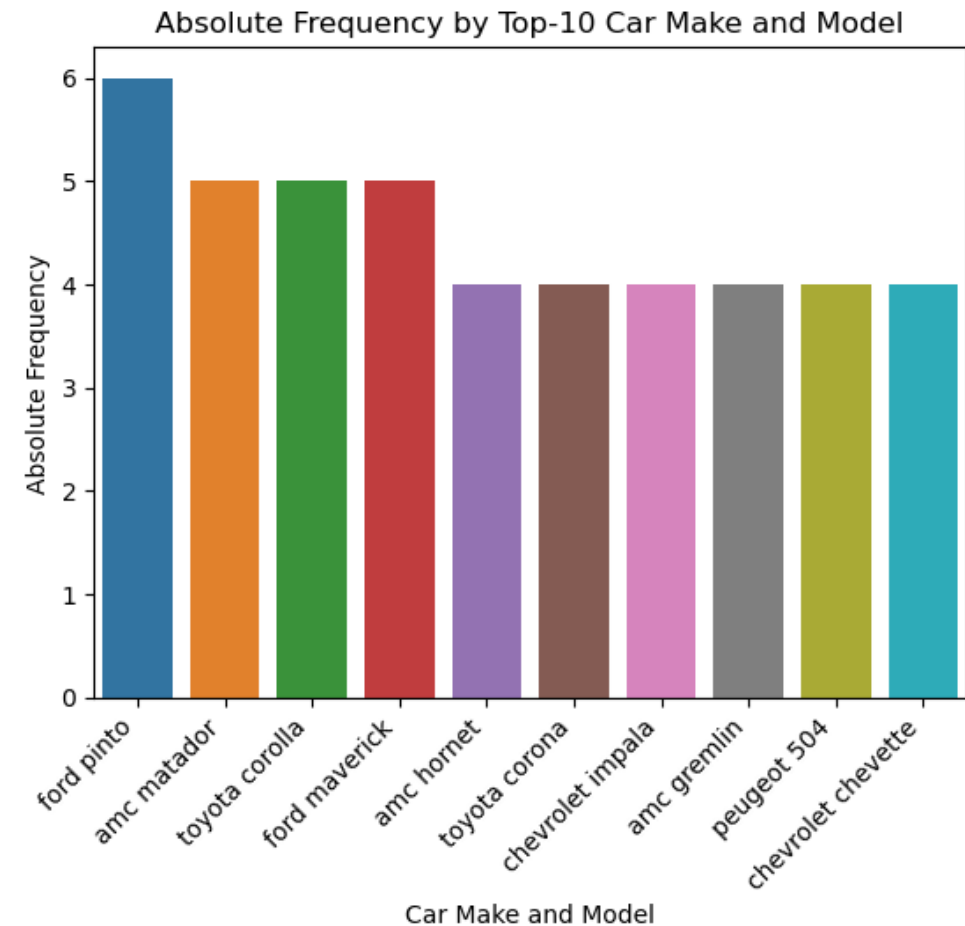
Cylinders	Absolute Frequency	Relative Frequency
4	202	0.51
8	103	0.26
6	84	0.21
3	4	0.01
5	3	0.01



EDA UNIVARIATE - CATEGORICAL

Top-10 car names

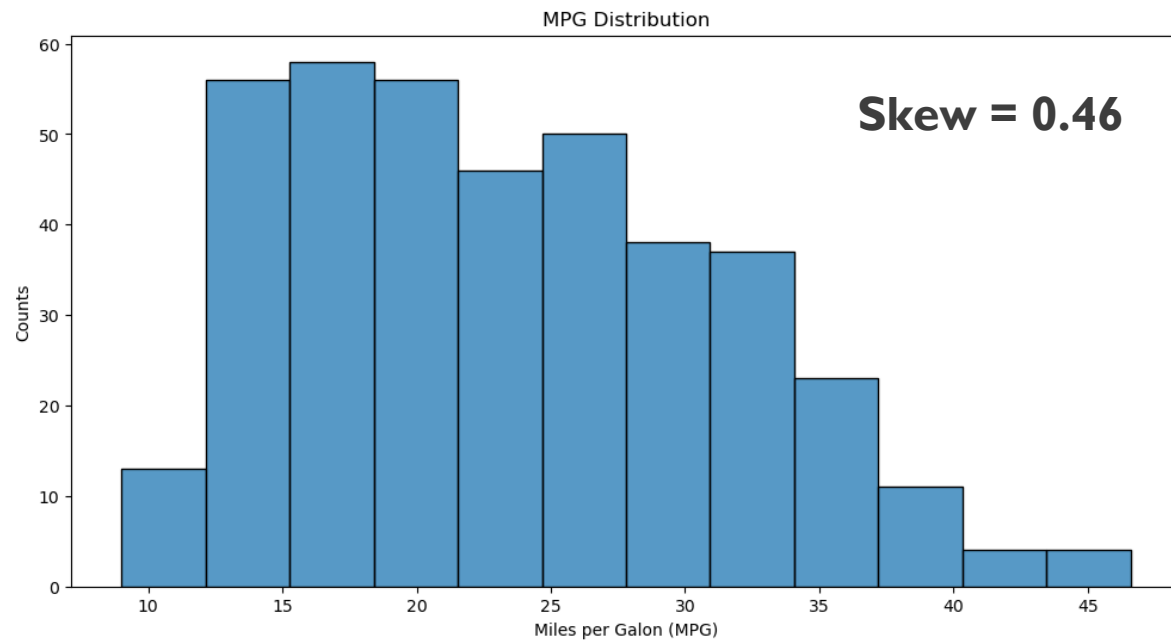
Origin	Absolute Frequency	Relative Frequency
ford pinto	6	0.02
amc matador	5	0.01
toyota corolla	5	0.01
ford maverick	5	0.01
amc hornet	4	0.01
toyota corona	4	0.01
chevrolet impala	4	0.01
amc gremlin	4	0.01
peugeot 504	4	0.01
chevrolet chevette	4	0.01



EDA UNIVARIATE - CONTINUOUS

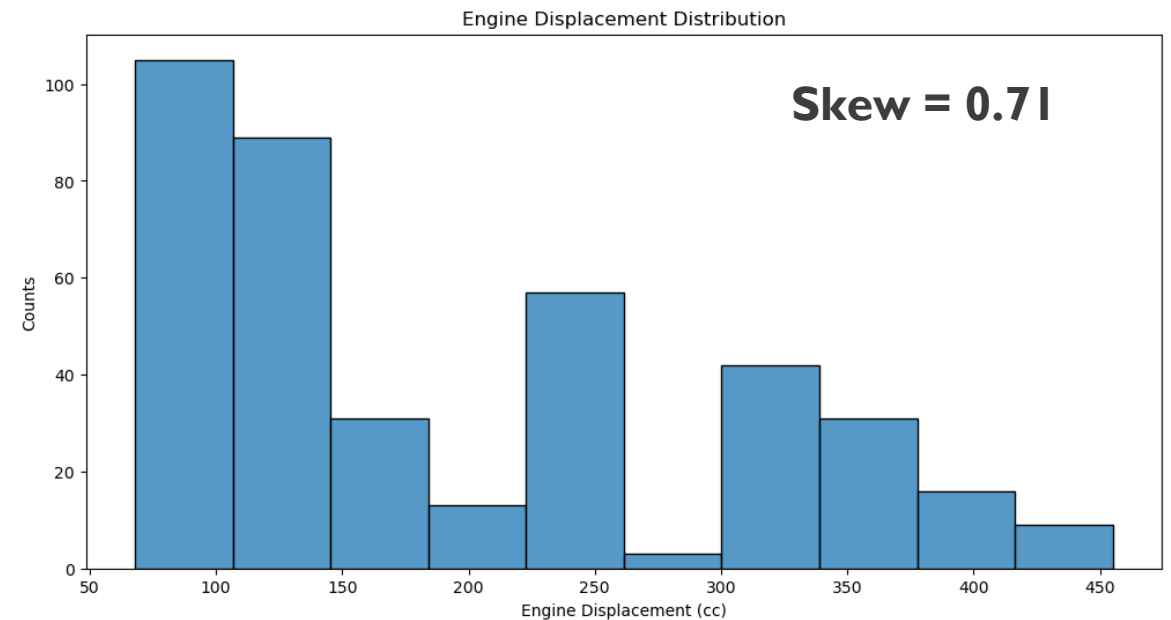
MPG

Mean	Std	Median	Mode
23.4	7.8	23.0	13.0



Displacement (cc)

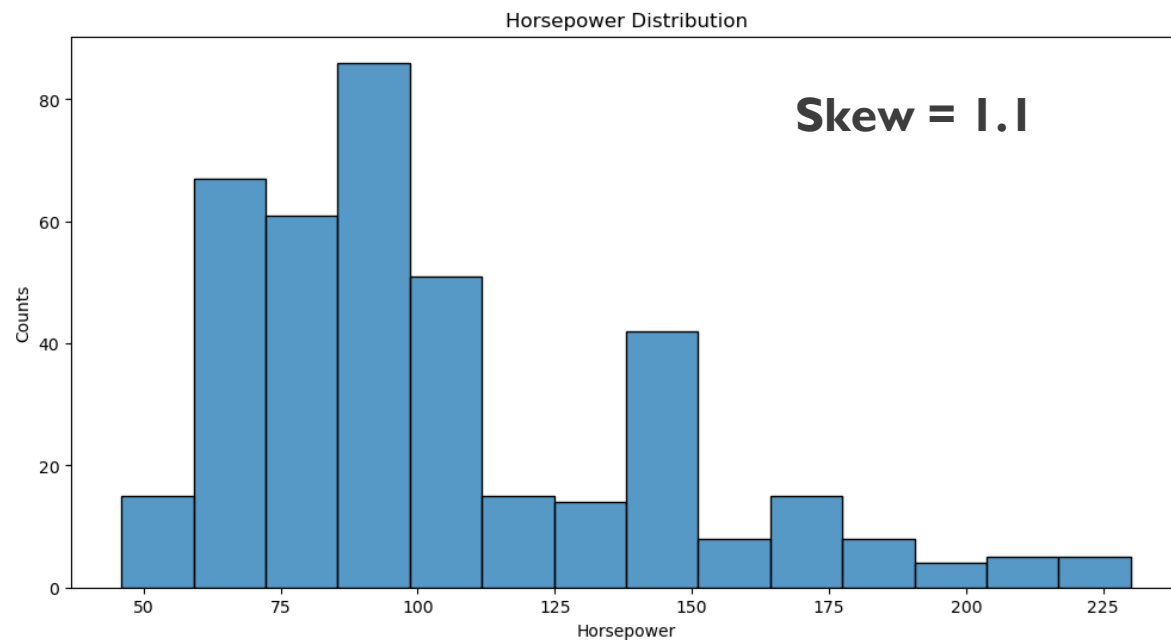
Mean	Std	Median	Mode
193.9	104.3	151.0	97.0



EDA UNIVARIATE - CONTINUOUS

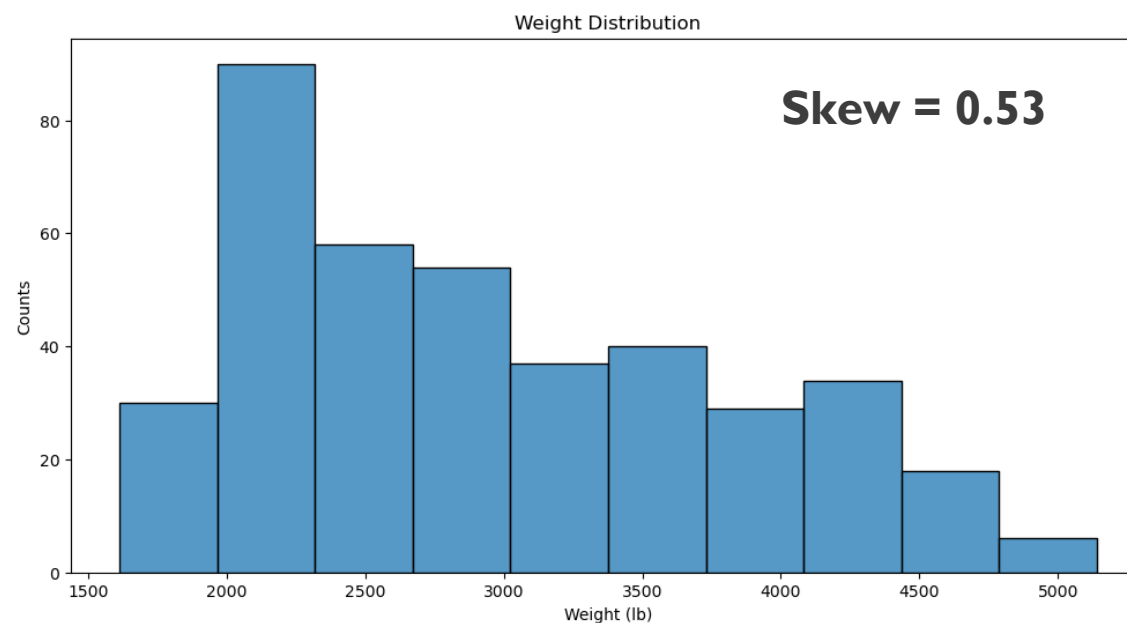
Horsepower

Mean	Std	Median	Mode
104.3	38.3	92.0	150.0



Weight (lb)

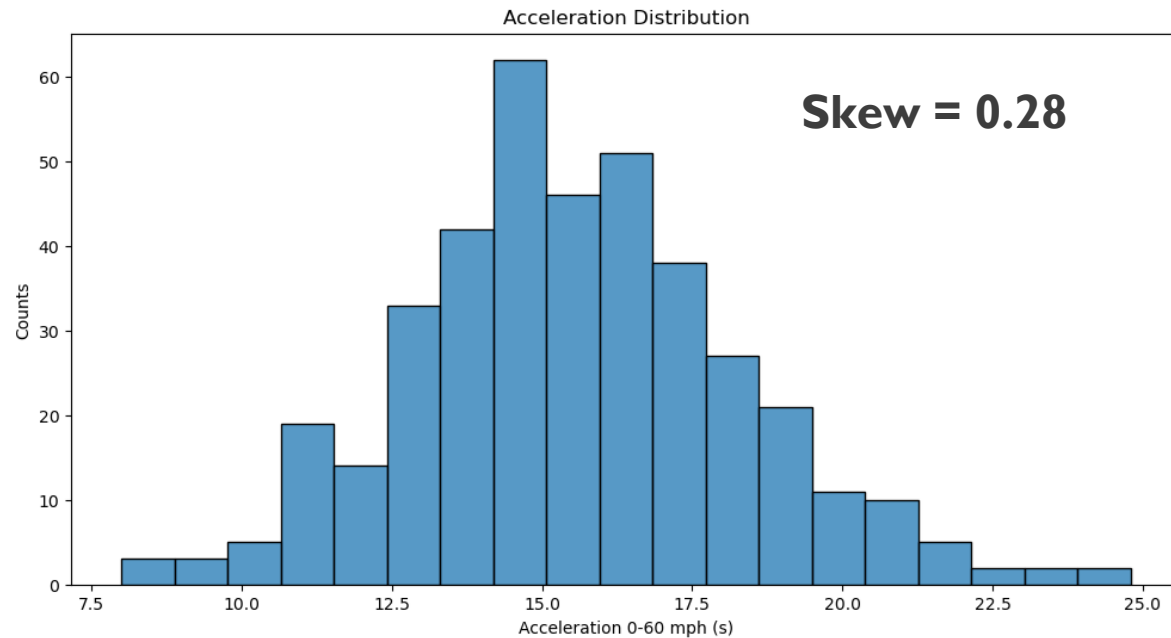
Mean	Std	Median	Mode
2974.9	846.4	2811.0	1985.0



EDA UNIVARIATE - CONTINUOUS

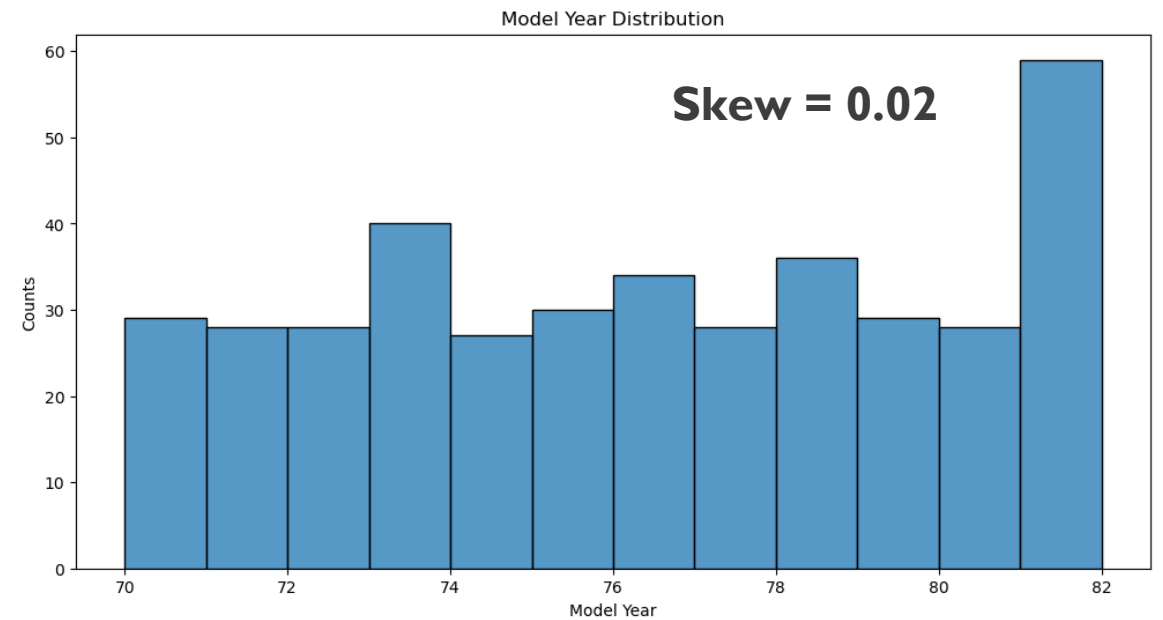
Acceleration 0-60 mph (s)

Mean	Std	Median	Mode
15.6	2.8	15.5	14.5



Model Year

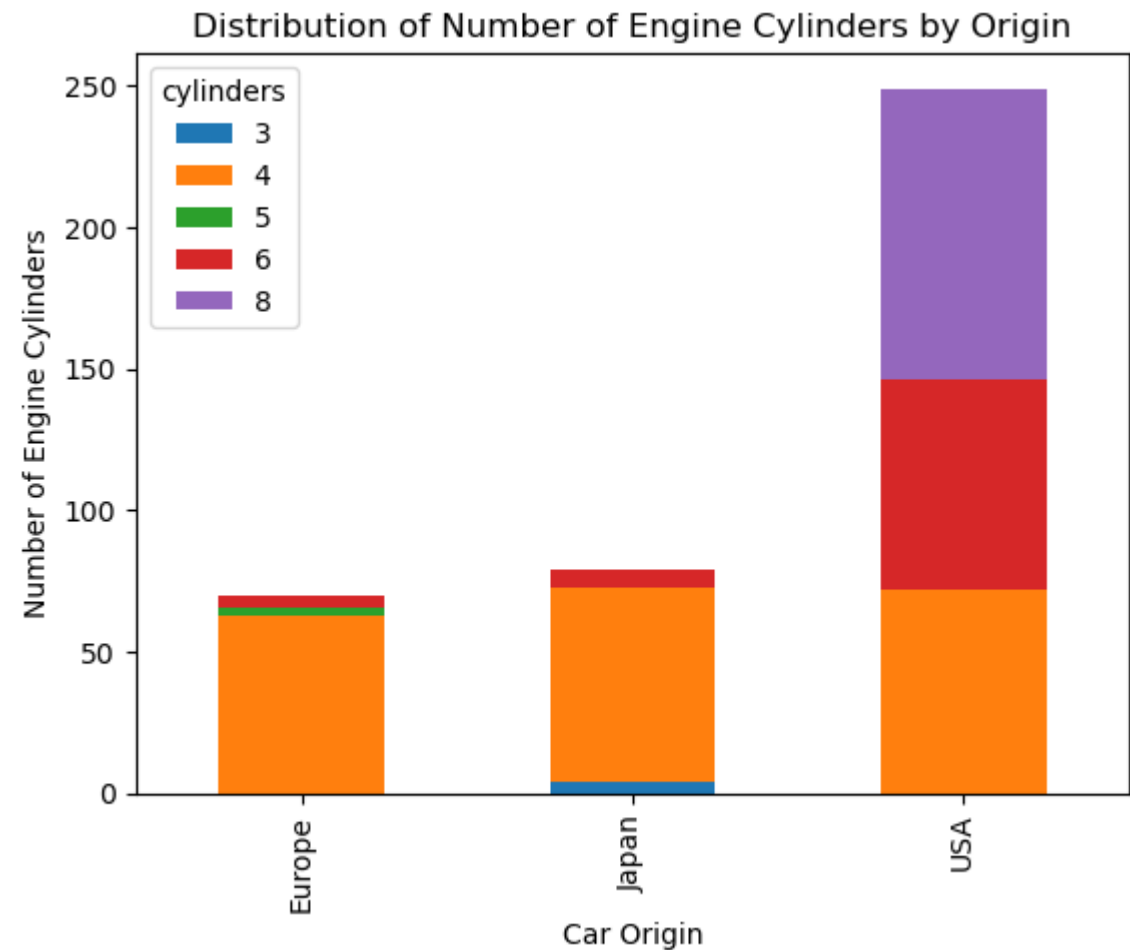
Mean	Std	Median	Mode
76.0	3.7	76.0	73.0



EDA BIVARIATE – CATEGORICAL VS CATEGORICAL

Origin vs Engine Cylinders

Origin	3	4	5	6	8
USA	0	72	0	74	103
Europe	0	63	3	4	0
Japan	4	69	0	6	0

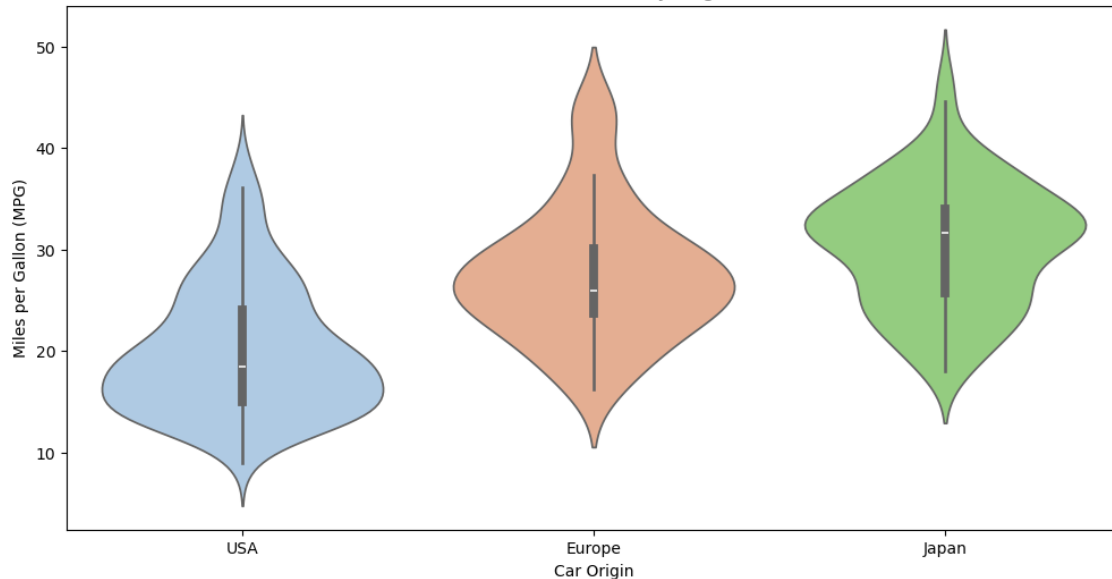


EDA BIVARIATE – CATEGORICAL VS CONTINUOUS

Origin vs MPG

Origin	Mean	Std	Median
USA	20.1	6.4	18.5
Europe	27.6	6.6	26.0
Japan	30.5	6.1	31.6

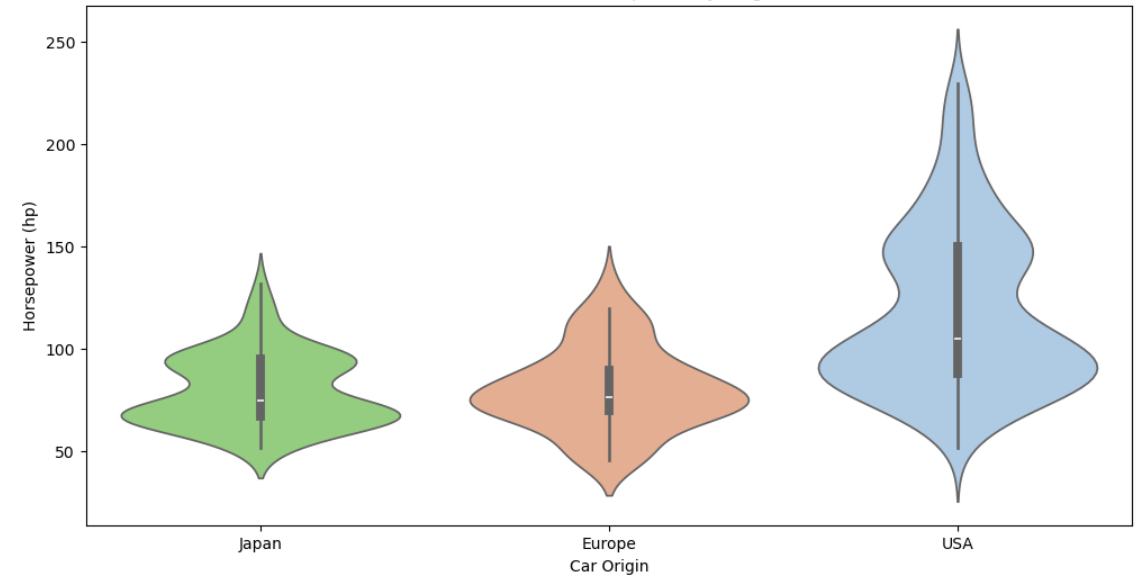
Distribution of MPG by Origin



Origin vs Horsepower

Origin	Mean	Std	Median
Japan	79.8	17.8	75.0
Europe	80.6	20.2	76.5
USA	118.5	39.8	105.0

Distribution of Horsepower by Origin

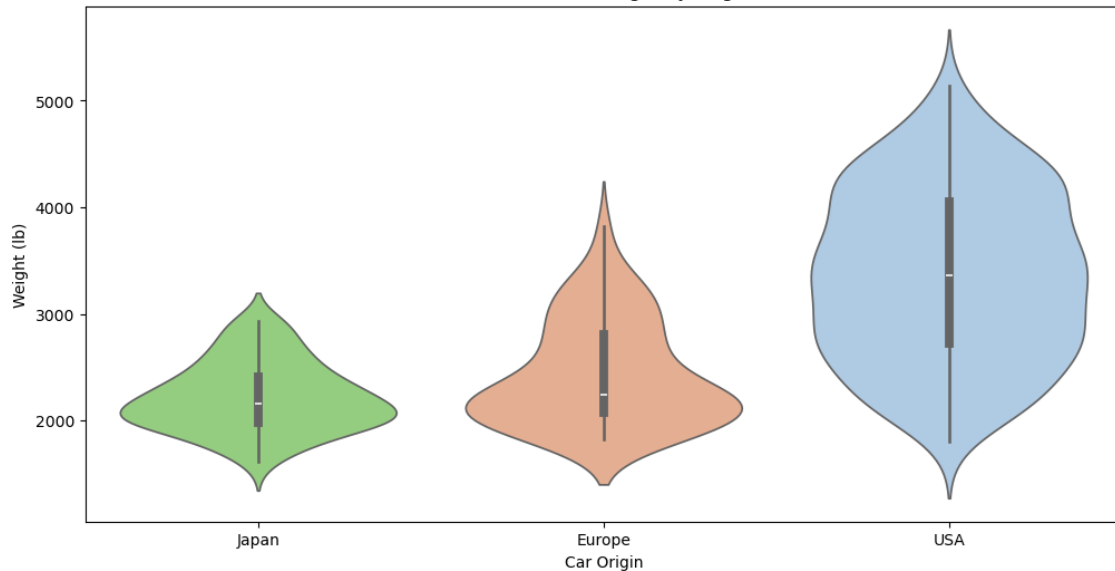


EDA BIVARIATE – CATEGORICAL VS CONTINUOUS

Origin vs Weight (lb)

Origin	Mean	Std	Median
Japan	2221.2	320.5	2155.0
Europe	2433.5	491.8	2240.0
USA	3361.9	794.8	3365.0

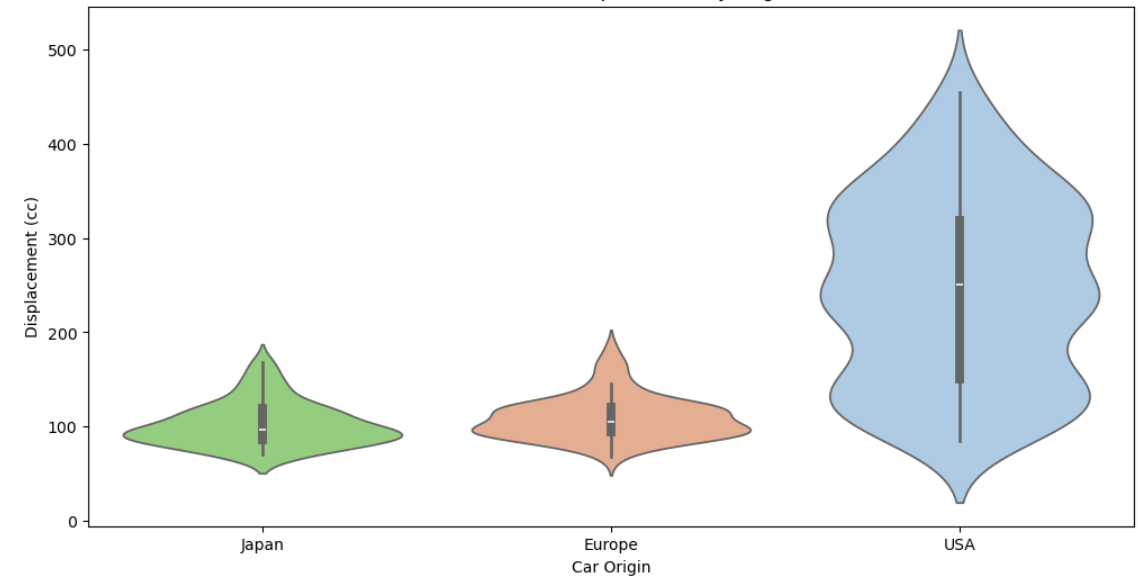
Distribution of Weight by Origin



Origin vs Displacement (cc)

Origin	Mean	Std	Median
Japan	102.7	23.1	97.0
Europe	109.6	22.7	105.0
USA	245.9	98.5	250.0

Distribution of Displacement by Origin

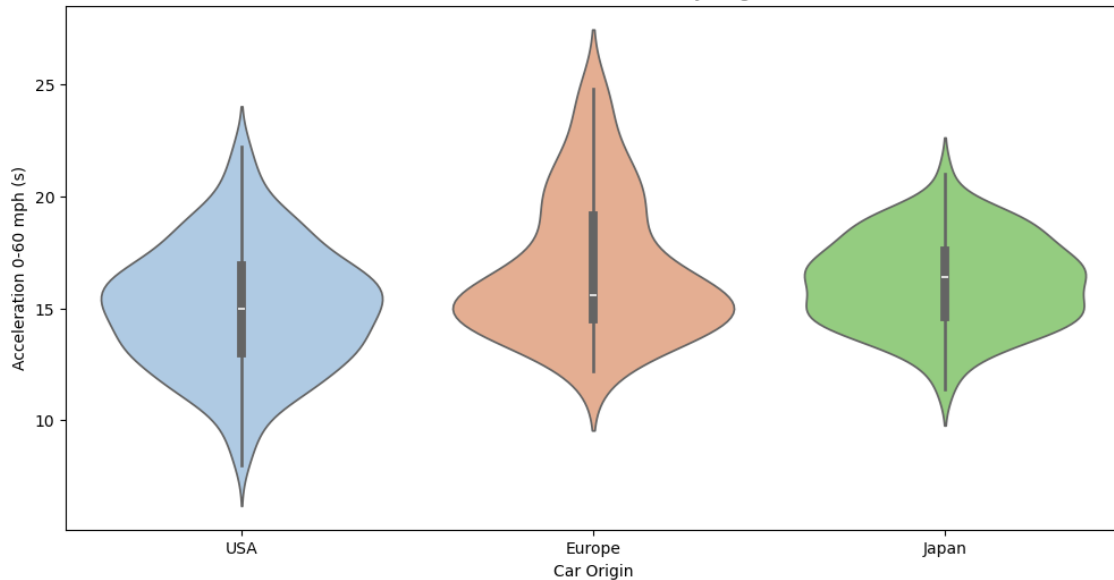


EDA BIVARIATE – CATEGORICAL VS CONTINUOUS

Origin vs Acceleration (s)

Origin	Mean	Std	Median
USA	15.0	2.8	15.0
Europe	16.8	3.1	15.6
Japan	16.2	2.0	16.4

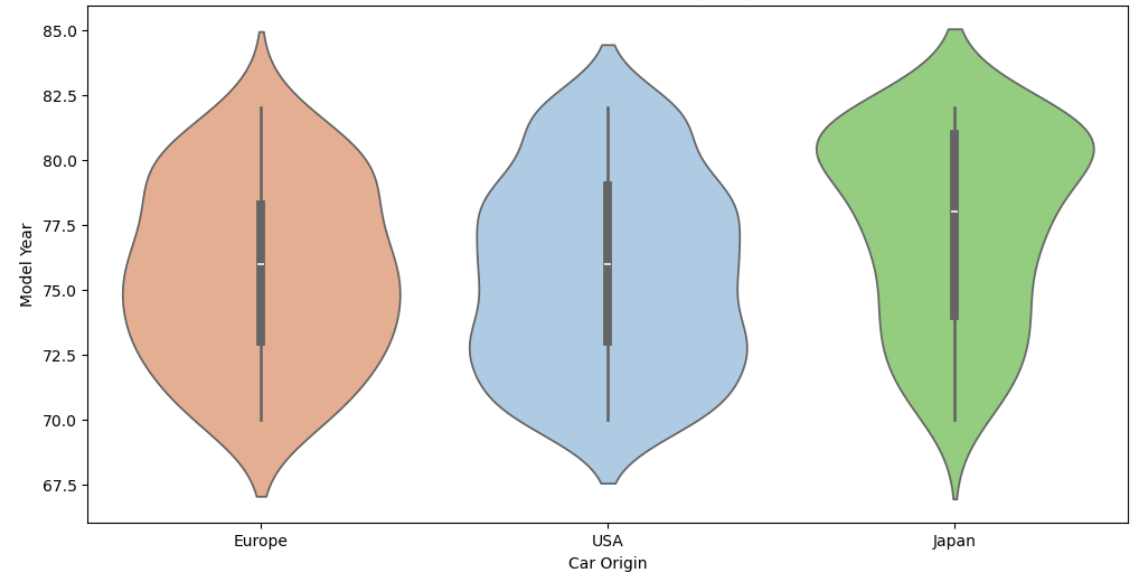
Distribution of Acceleration by Origin



Origin vs Model Year

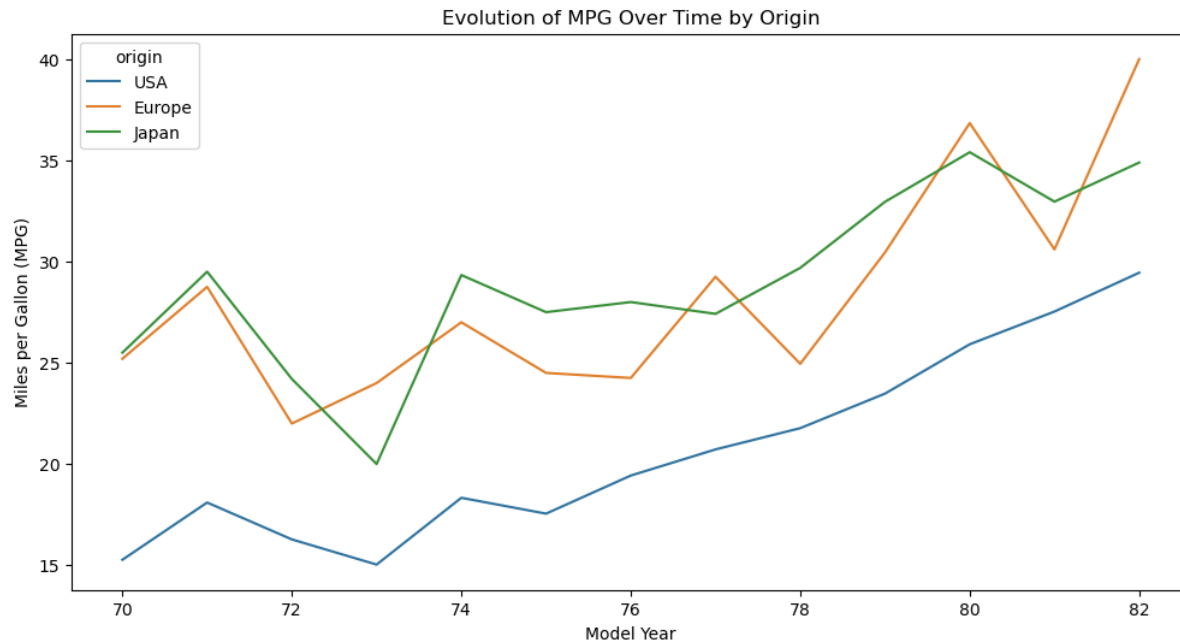
Origin	Mean	Std	Median
Europe	75.7	3.4	76.0
USA	75.6	3.7	76.0
Japan	77.4	3.7	78.0

Distribution of Model Year by Origin

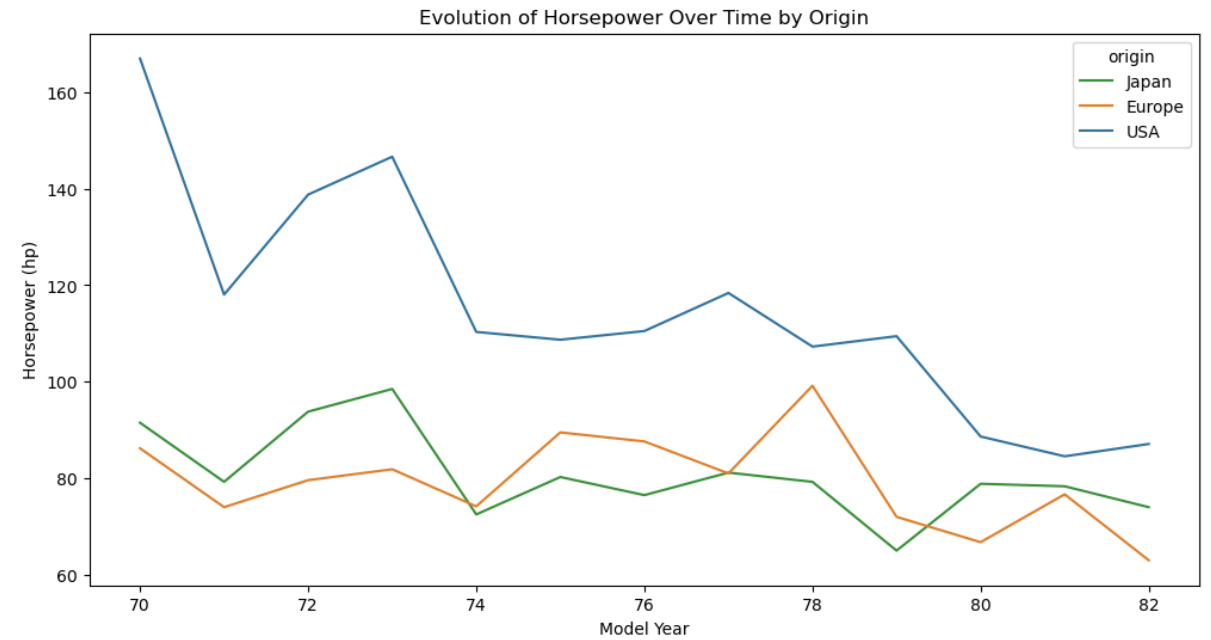


EDA MULTIVARIATE

Evolution of MPG Over Time by Origin

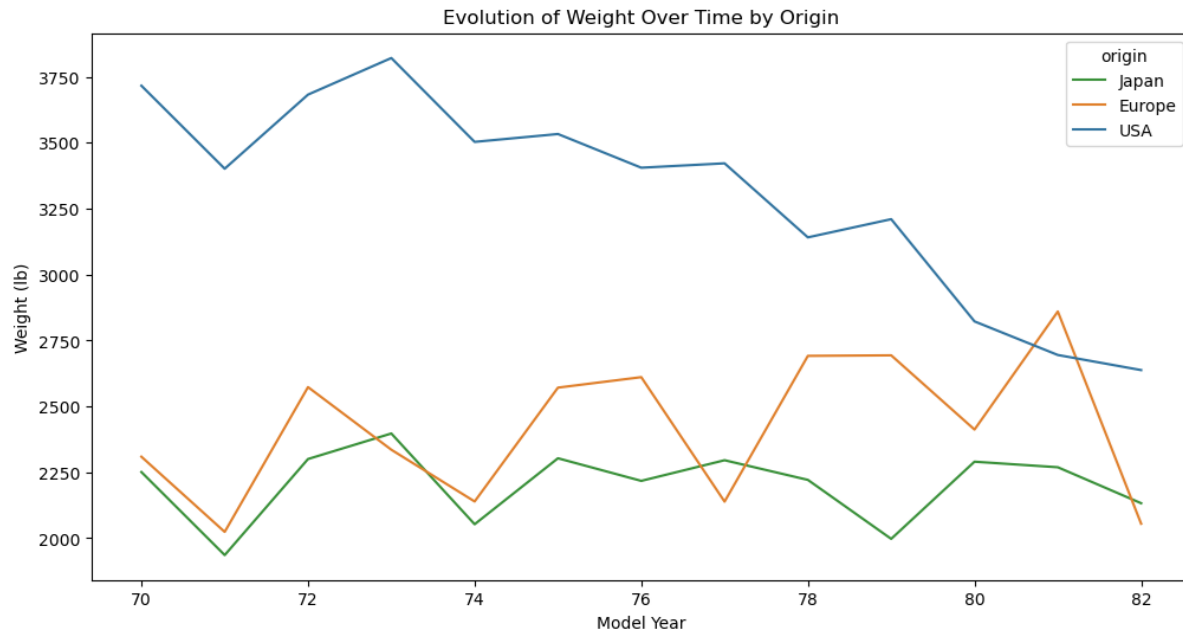


Evolution of Horsepower Over Time by Origin

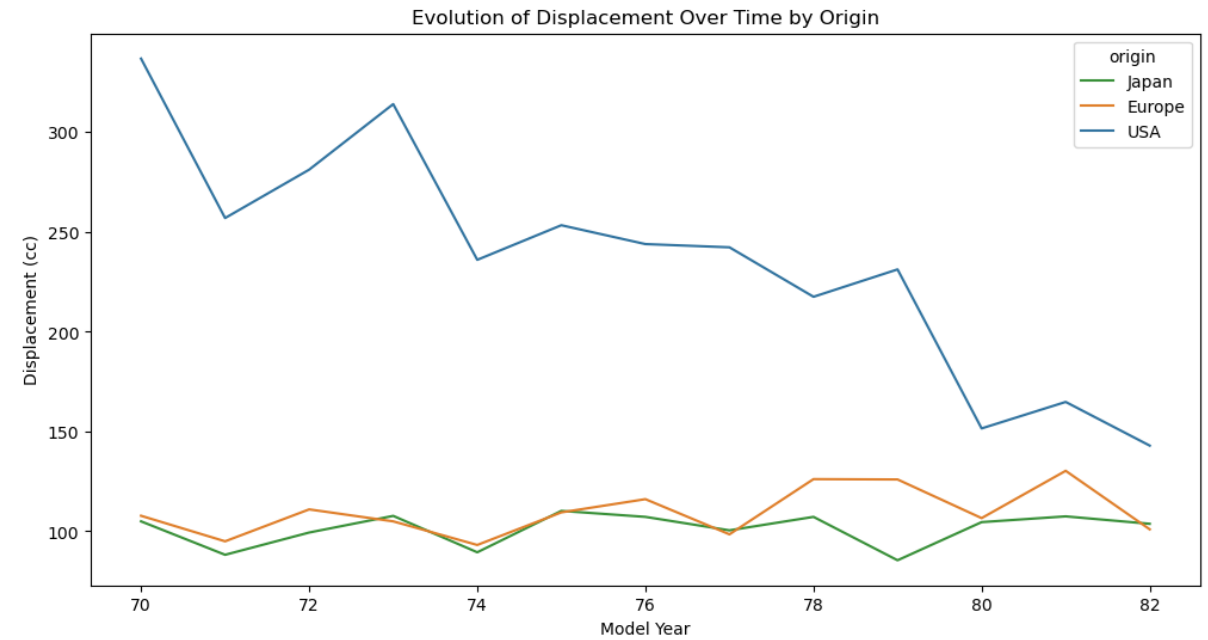


EDA MULTIVARIATE

Evolution of Weight Over Time by Origin

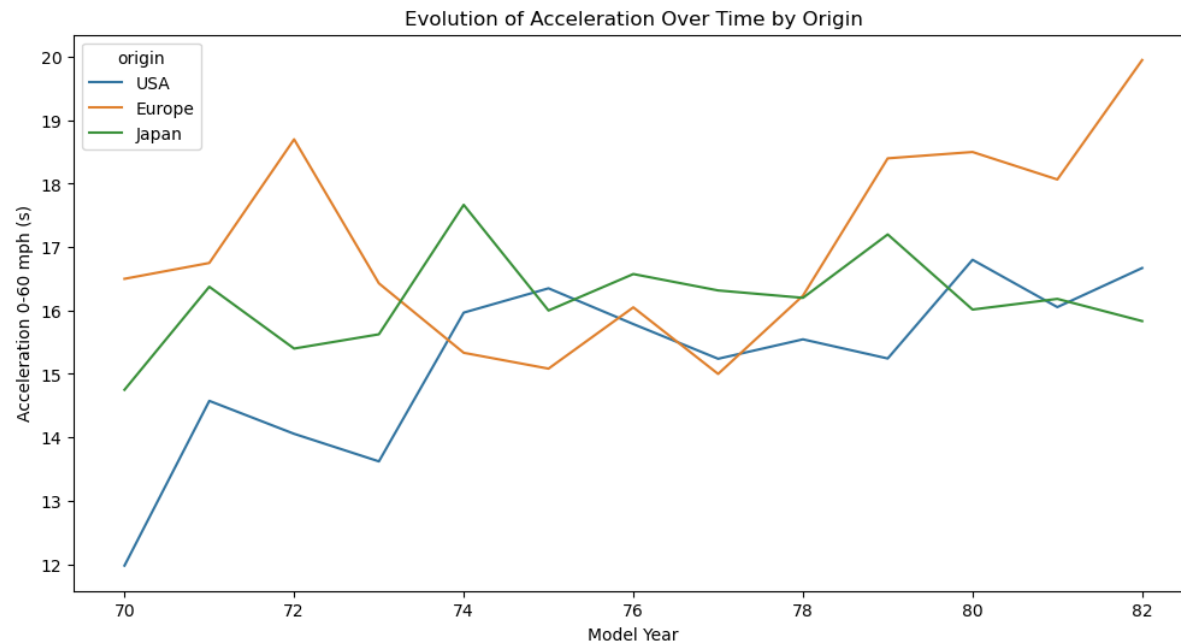


Evolution of Displacement Over Time by Origin



EDA MULTIVARIATE

Evolution of Acceleration Over Time by Origin



KEY INSIGHTS AND CONCLUSIONS

Univariate EDA

- Dataset dominated by USA cars (~63%); mostly 4-cylinder engines;
- Most variables show right-skewed distributions (MPG, horsepower, weight, displacement).

Bivariate EDA

- Fuel efficiency increases: USA → Europe → Japan;
- USA cars: higher horsepower, weight, variability;
- Europe and Japan: lighter, more efficient, more consistent.

Multivariate EDA (Time Evolution)

- Fuel efficiency improves over time across all regions;
- USA shows convergence but remains less efficient;
- USA shows decrease in horsepower, weight, displacement;
- Japan and Europe remain more stable and efficient.

For the car population under analysis, in the period between 1970 and 1982 car origin strongly reflected different engineering philosophies, with a clear shift over time towards efficiency across all regions.

CHALLENGES AND SOLUTIONS

Missing Data (hp - 6 cases)

- Used domain-based imputation (similar cars by engine cylinder configuration);
- Dropped cases where no reliable estimate was possible.

Outliers (hp, MPG and acc.)

- Used IQR (Tukey method) for detection;
- Decided to retain original data to preserve real-world variability;
- Outliers could represent meaningful sub-populations (more context required).

Variable Interpretation

- Numerical vs categorical ambiguity (e.g.: cylinders, origin);
- Reclassified variables based on analytical relevance.

Data Handling and Indexing

- Issues when dropping rows and managing indices;
- Preserved index consistency;
- Avoided unnecessary resets to maintain data integrity.

FUTURE WORK/NEXT STEPS

Further Analysis

- Power-to-weight ratio analysis by origin;
- Detailed analysis of efficiency trends over time;
- Year-on-year improvement rates.

Advanced Methods

- Statistical testing (significance of observed trends);
- Deeper segmentation of car populations.

SQL Extension

- Translate insights into targeted queries;
- Average MPG per year;
- Efficiency by origin over time.



Thank You!